NOISEMAP · USER REQUIREMENTS

NoiseMap user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue (kit 1.8 pipeline); drawn from the users and context in NSM-PRB-001 and the existing requirements

NoiseMap user requirements

What the people who would live with, buy, fit or study the node need, written from their side. The product requirements that meet these needs are in the product requirements (`03-requirements.md` , NSM-PRS-001) and the links to evidence are in the traceability matrix (`03b-traceability.md`). Sources are the user types and stakeholders in NSM-PRB-001 (Tables 1 and 2). No co-design partner has yet reviewed these needs (NSM-PRB-001 open items), so all priorities are proposals, "Proposed, awaiting Amish".

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Regulatory	No eavesdropping	The people on the street can be sure no audio of them is stored or leaves the device; only sound levels do.	High	Residents and associations; venue operators (PRB Table 1)
UR-002	Performance	Night-time evidence	A continuous record of how loud it was and when, at the street, in a form a council or licensing panel can use (LAeq, peaks, a low-frequency measure).	High	Residents; city environmental health team; licensing authority
UR-003	Performance	Credible accuracy	Readings close enough to a class 1 reference meter to be trusted at a hearing, with the uncertainty stated.	High	City environmental health team; licensing authority; researchers
UR-004	Performance	Quiet to loud	The node resolves a quiet residential night and a loud motorcycle or siren.	High	Residents; city environmental health team
UR-005	Performance	Bass from venues	Levels that reflect low-frequency amplified music as well as A-weighted traffic noise.	Medium	Residents; licensing authority; venue operators
UR-006	Performance	Events can be dated	Each record carries a trusted time so a complaint can be matched to a level.	Medium	Licensing authority; residents
UR-007	Deployment and assembly	No power at the pole	The node runs for years without mains power or visits, through dull weeks.	High	Pole or asset owner; city environmental health team
UR-008	Deployment and assembly	Clamp on, nothing drilled	Two trained people fit it to an existing pole in one short visit with no drilling, welding or wiring into the pole.	High	Pole or asset owner
UR-009	Safety	Light and wind-proof	The load on the pole stays small, the node stays put in a storm, and a falling part cannot land on a passer-by.	High	Pole or asset owner; public
UR-010	Environment and durability	Outdoors for years	The node works in sun, rain, dust and -20 to +50 degrees C for about five years.	Medium	City environmental health team; pole owner
UR-011	Usability	Know which data are spoiled	Records made in wind or heavy rain are marked so analysts can leave them out.	Medium	Researchers; city environmental health team
UR-012	Maintenance	Quick field check	A technician can check the node against a calibrator in a few minutes at the pole.	Medium	City environmental health team; researchers
UR-013	Regulatory	Shares the airwaves fairly	Radio use stays within the licence-free band rules and shared network fair use.	Medium	City environmental health team; network operators
UR-014	Cost			High	

ID	Category	Title	Need	Priority	Source
		Cheap enough for many streets	The parts for one node are affordable for a residents' group or a small city team.		Residents' associations; city environmental health team
UR-015	Regulatory	Open and checkable	Hardware, firmware, calibration method and data format are published so anyone can audit what is measured.	High	Residents; researchers; city teams
UR-016	Regulatory	Public notice	Passers-by can read on the pole what is measured and where to find the design.	Medium	Residents; venue operators; pole owner

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.